MACAU UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Computer Science and Engineering

Faculty of Innovation Engineering

Software Project for Course Software Engineering

Homework ID: Task2 - User Requirements Specifications

Report Title: Campus Second-hand Trading Platform

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Amendment History

Version	Date	Modified By	Reviewed By	Amendment Details
1.0	2026/10/18	Group 11 (SeekDeep)	Group 11 members	Initial version with the full feature set (incl. automatic matching, want-to-buy requests and notifications).
1.1	2026/10/19	Group 11	-	Reduced project scope to an achievable student prototype; corrected the requirement-elicitation section.
1.2	2026/10/20	Group 11	-	Updated the elicitation record after Group 10 feedback; clarified deletion-related data handling (REQ-1.3, REQ-2.2); kept automatic matching outside version 1.
1.3	2026/10/20	Group 11	-	Merged the complete template structure into the reduced v1.2 scope: restored Notation, Context Diagram, per-requirement Display and Data Handling fields, and the Availability and Capacity sections.

1. Introduction

1.1 Purpose

This document is the User Requirements Specification (URS) for the Campus Second-hand Trading Platform, a web-based course project for students of the Macau University of Science and Technology (MUST). It specifies the functional, non-functional and domain requirements of the system from the users' point of view. It builds on the Task1 Project Proposal and will serve as the input for the High-Level Design (HLD) and Low-Level Design (LLD) in the following tasks.

The specification intentionally focuses on a basic and complete second-hand trading workflow rather than advanced functions such as AI recommendation, online payment, delivery logistics or complex real-time services.

1.2 Notation

This specification and its supporting documents and diagrams use the Unified Modelling Language (UML) notation. Requirement identifiers follow the pattern REQ-x.y for functional requirements, PERF/SEC/AVA/CAP for non-functional requirements, and DOM for domain requirements.

1.3 Scope

The first version of the system supports the core workflow of campus second-hand trading: account registration and login, publishing and managing product listings, browsing/searching/filtering products, saving favorites, viewing seller-provided contact information, and changing product status between Available, Reserved and Sold.

Out of scope for version 1:

- Online payment and refund processing.
- Delivery or logistics management.
- AI recommendation, want-to-buy automatic matching, buyer-seller matching rules, and machine-learning models.

Built-in real-time chat, e-mail/SMS notification, and external identity verification.

Scope change from the Task1 proposal: the Task1 proposal identified automatic buyer-seller matching (F5) as the core value of the system. After feedback from the assigned customer group (Group 10) on 2026/10/20, automatic matching, want-to-buy requests and in-system notifications were removed from version 1 (see Section 10.1). The system is therefore deliberately scoped to a basic but complete second-hand trading workflow that is achievable and fully testable within the course. Seller contact information (REQ-6.1) replaces automatic matching so that buyers and sellers can still reach each other and complete the trade offline.

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1.4 Context Diagram

The system is a standalone web application. It interacts with the following external actors and systems:

- Students (buyers and sellers): use the system through a web browser.
- Group 10 (assigned customer group): provides requirements and feedback as the client of Group 11.
- MySQL 8 database: stores all persistent data, hosted on the same server.

Payment, negotiation and physical handover take place outside the platform between the students themselves. Figure 1 shows the context diagram of the system: the platform appears as a single node exchanging data with its external actors and systems.

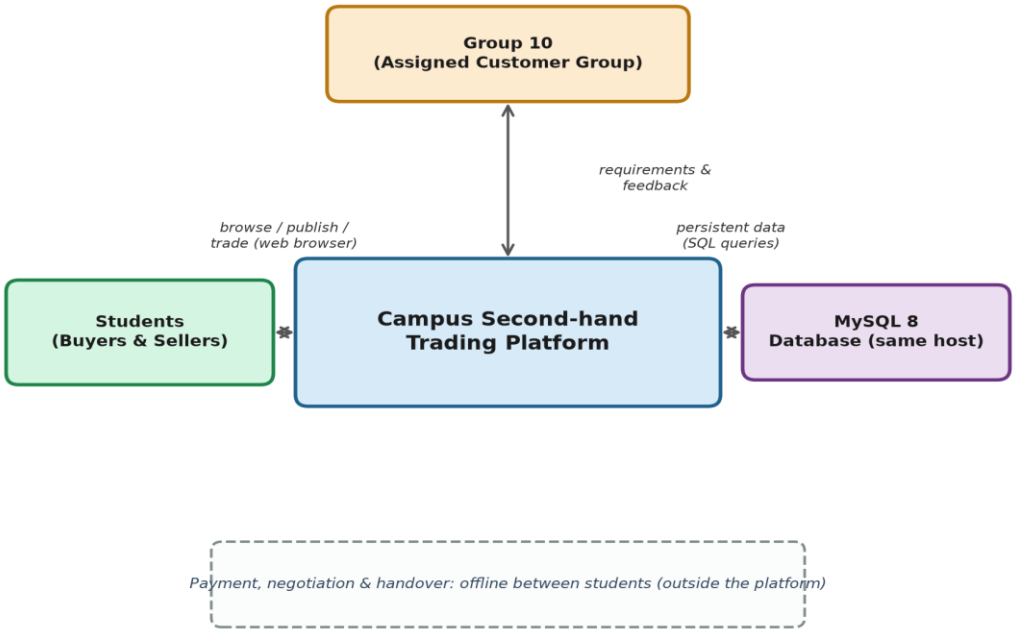


Figure 1: Context diagram of the Campus Second-hand Trading Platform

1.5 Definitions and Acronyms

Term	Definition
Listing	A second-hand product posted by a seller, with title, category, description, condition and price.
Available / Reserved / Sold	Transaction status values of a listing.
Favorites	A user's saved list of listings.
MOP	Macau Pataca, the currency used for all prices in the system.
P95	The 95th percentile response time.
UML	Unified Modelling Language.
URS / SRS	User / Software Requirements Specification.
MUST	Macau University of Science and Technology.

1.6 References

[1] Task1 Project Proposal - Campus Second-hand Trading Platform, Group 11.

[2] Task2 - User Requirements Specifications Description, Software Engineering course, MUST.

[3] Task2 User Requirements Specifications Template, Moodle platform.

[4] Project repository: https://github.com/cccc1ccc/smart-campus-secondhand

1.7 Overview

Section 2 gives a general description of the system functions and user types. Sections 3 and 4 list general constraints, assumptions and dependencies. Sections 5 and 6 define the functional requirements in detail (REQ-1.1 to REQ-6.1). Section 7 covers external interfaces. Section 8 covers the non-functional requirements (performance, security, availability, capacity). Section 9 states the domain requirements. Section 10 records the requirement elicitation process with the three sources required by the course, and Section 11 lists the contribution of each member to this report.

2. General Description

2.1 System Functions

The system provides seven high-level functions, inherited from the Task1 proposal and confirmed by requirement elicitation:

Function Name	Summary
F1 User Registration and Login	Students can create an account and log in.
F2 Profile and Account Management	Users can maintain basic profile information and delete their own account.
F3 Product Listing Management	Sellers can publish, edit and delete their own listings.
F4 Browse, Search and Filter	Buyers can browse listings and search/filter by simple conditions.
F5 Favorites	Users can save and remove listings from a favorites list.
F6 Transaction Status Management	Sellers can change listing status to Available, Reserved or Sold.
F7 Seller Contact Information	A buyer can view the contact method voluntarily provided by the seller.

2.2 User Characteristics

User Type	Characteristics	Interface
Seller	MUST student who wants to sell used items (textbooks, electronics, household items). Occasional user, no training needed.	Web browser
Buyer	MUST student looking for cheaper second-hand items. Occasional user, no training needed.	Web browser
Both roles	The same student account can act as seller and buyer at the same time; no separate registration is required.	Web browser

All users are expected to read English or Chinese; version 1 provides an English user interface. No data-entry training is needed.

3. General Constraints

3.1 Software Constraints

- The system will be implemented as a web application with Java 17 and Spring Boot, using Maven as the build tool and JUnit 5 for unit testing (course-approved stack fixed in Task1).
- Persistent data must be stored in MySQL 8.
- Version 1 needs to support current desktop versions of Chrome and Edge; mobile-friendly layout is desirable but not mandatory for every page.
- Source code will be hosted on the project GitHub repository.
- Only free or open-source libraries may be used; no licence cost is allowed.

3.2 Hardware Constraints

- The system must run on a single ordinary development computer or a low-cost/free cloud host; no dedicated server hardware is available.
- Users need only a normal laptop or phone with a web browser and a network connection.

4. Assumptions and Dependencies

- A1: Users are MUST students who can access the website through a browser.
- A2: Payment, price negotiation and physical handover are completed outside the system; the system only records the transaction status.
- A3: Sellers voluntarily provide any contact information shown to buyers.
- A4: The first version is a course prototype and is not intended for university-wide production deployment.
- D1: Persistent data depends on the selected relational database (MySQL 8) and a Java 17 runtime on the host machine.
- D2: The performance benchmark depends on a seeded database of at least 5,000 listings kept in the project repository.

5. Functional Requirements Master List

The table below lists all functional requirements in summary form. Each requirement is defined in detail in Section 6.

Req. ID	Requirement Name	Requirement Description
REQ-1.1	User registration	A visitor can create an account with basic information.
REQ-1.2	User login/logout	A registered user can log in and log out.
REQ-1.3	Profile and account management	A user can edit basic profile information or delete his or her own account.
REQ-2.1	Publish listing	A seller can publish a listing with basic product information.
REQ-2.2	Manage listing	A seller can edit/delete his or her own listing; related favorites are cleaned up when a listing is deleted.

REQ-3.1	Browse/search/filter	A buyer can browse, search and apply simple filters.
REQ-4.1	Favorites	A user can add/remove a listing from favorites.
REQ-5.1	Transaction status	A seller can change a listing between Available, Reserved and Sold.
REQ-6.1	Seller contact	A buyer can view seller contact information when the seller chooses to provide it.

6. Functional Requirements (Detailed)

6.1 Functional Requirement REQ-1.1 User Registration

Description: A visitor creates a basic user account before publishing or saving listings.

System Input: Display name, student number or e-mail, and password.

Display: A registration form with clearly labelled fields, inline validation messages and a submit button.

System Processing: The system checks that required fields are not empty and that the account identifier is not already used. The password is stored securely as a salted hash.

System Output: The new account is created and the user can proceed to log in.

Other: None.

Constraints: Registration is limited to a small number of basic fields.

Data Handling: All strings are stored in UTF-8 (Unicode); the account identifier and e-mail are validated before the account is persisted.

Error Handling: Duplicate account information or invalid input is rejected with a clear message.

6.2 Functional Requirement REQ-1.2 User Login and Logout

Description: A registered user can log in to use account-based functions and can log out after use.

System Input: Account identifier and password.

Display: A login form with clearly labelled fields; a logout control is visible while logged in.

System Processing: The system verifies the credentials against the stored hash and creates a user session after successful login.

System Output: Successful login opens the home page; logout ends the current session.

Other: The session expires after 2 hours of inactivity.

Constraints: None.

Data Handling: Credentials are verified against the stored salted hash and are never compared or logged in plain text.

Error Handling: Wrong credentials show a generic login failure message, without hinting which field failed.

6.3 Functional Requirement REQ-1.3 Profile and Account Management

Description: A user can edit simple profile information and may delete his or her own account.

System Input: Display name, optional contact method, password-change fields, or an account-deletion confirmation action.

Display: A profile page showing the current data with edit capability.

System Processing: The system validates profile changes before saving them. If the user confirms account deletion, the system deletes the user profile, the user's own listings and the user's favorites. Favorites created by other users for listings owned by the deleted account are also removed because those listings no longer exist.

System Output: The updated profile is displayed, or the user is logged out after successful account deletion.

Other: None.

Constraints: The user controls whether contact information is provided. Account deletion requires explicit confirmation. Version 1 has no matching records or notification history to preserve because automatic matching and notification services are outside scope.

Data Handling: Profile changes are validated before saving; a password change requires the old password and the new password must differ from the old one.

Error Handling: Invalid profile data is rejected without changing the existing profile. A failed account deletion does not partially remove user data.

6.4 Functional Requirement REQ-2.1 Publish Listing

Description: A logged-in seller can publish a second-hand product.

System Input: Title, category (from a fixed list), description, condition (new / like-new / good / fair), asking price in MOP, and one optional image.

Display: A publishing form with clearly labelled fields (at most five mandatory fields), inline validation messages and a submit button.

System Processing: The system validates the required fields and the price as a positive number, saves the listing with status Available and the current timestamp, and stores the optional image if provided.

System Output: The new listing page is displayed and the item becomes visible in browsing/search results.

Other: Photos are optional in version 1 (one photo per listing, JPEG/PNG, at most 5 MB).

Constraints: Only logged-in users can publish. Image upload is optional. A user can have at most 50 active listings.

Data Handling: Title is limited to 100 characters, description to 2,000 characters, and the price is stored as a number with two decimal places in MOP.

Error Handling: Missing required fields or invalid price values are reported to the user with the offending field highlighted. An oversized photo is rejected with a size hint.

6.5 Functional Requirement REQ-2.2 Manage Listing

Description: A seller can edit or delete his or her own listing.

System Input: A listing selected from My Listings and the fields to update.

Display: A 'My Listings' page showing the seller's listings with their status, and an edit page identical in layout to the publish form.

System Processing: The system checks ownership before allowing an update or deletion. When a listing is deleted, all favorites that reference that listing are also removed so that users do not keep invalid saved records.

System Output: The listing is updated or removed and a confirmation message is shown. A deleted listing disappears from search results and favorites.

Other: None.

Constraints: A user cannot edit or delete another user's listing.

Data Handling: Every update is timestamped; the edit history is not kept in version 1.

Error Handling: Unauthorized operations are rejected with an explanatory message.

6.6 Functional Requirement REQ-3.1 Browse, Search and Filter

Description: A buyer can find products without using advanced recommendation algorithms.

System Input: Optional keyword, category, minimum and maximum price, condition, and sort order (newest, price ascending, price descending).

Display: A result list showing title, price, condition and status, grouped by page (20 items per page); applied filters are shown as removable chips.

System Processing: The system queries listings and returns items that satisfy the selected conditions. Keyword matching covers title and description. Sold listings are hidden from normal results.

System Output: The filtered and sorted result list; selecting an item opens its detail page. An empty-result message is shown when nothing matches.

Other: None.

Constraints: Simple database query and pagination are sufficient for version 1.

Data Handling: Numeric filters are validated; a minimum price greater than the maximum price is rejected.

Error Handling: Invalid price ranges or filter combinations are rejected with a clear message while the previous results are kept.

6.7 Functional Requirement REQ-4.1 Favorites

Description: A logged-in user can save interesting listings for later viewing.

System Input: Listing selected through an Add to Favorites / Remove from Favorites action.

Display: A 'Favorites' page listing all saved products with price and status, most recently saved first.

System Processing: The system adds or removes the relationship between the user and the listing. Duplicate saves of the same listing are ignored.

System Output: The favorites page shows the user's saved listings and a confirmation of the add/remove action.

Other: None.

Constraints: A user can save at most 200 favorites.

Data Handling: Favorites reference the listing ID only; no listing data is duplicated.

Error Handling: If the listing no longer exists, it is removed from the favorites view.

6.8 Functional Requirement REQ-5.1 Transaction Status Management

Description: The seller can indicate whether an item is Available, Reserved or Sold.

System Input: Listing and selected target status.

Display: The status is displayed as a coloured badge on the listing page, in search results, in favorites and in My Listings.

System Processing: The system checks that the current user owns the listing and saves the selected status. Allowed transitions: Available to Reserved, Reserved to Sold, and Reserved back to Available.

System Output: The new status is shown on the listing and in My Listings.

Other: None.

Constraints: Only the seller can change the status. Sold items do not appear in normal search results. Invalid transitions (e.g. Sold to Available) are not allowed in version 1.

Data Handling: Status history (timestamped transitions) is stored for the lifetime of the listing.

Error Handling: Unauthorized or invalid status changes are rejected with an explanatory message.

6.9 Functional Requirement REQ-6.1 Seller Contact Information

Description: A buyer can view a simple contact method when the seller chooses to provide one.

System Input: No extra buyer input; contact information is read from the seller profile or listing.

Display: The seller-provided contact information is shown on the listing detail page.

System Processing: The system displays the seller-provided contact information on the listing detail page.

System Output: The buyer can contact the seller outside the platform to discuss price and handover.

Other: None.

Constraints: The platform does not provide built-in chat in version 1.

Data Handling: Contact information is optional and is displayed only when the seller chooses to provide it.

Error Handling: If no contact information is provided, the page shows that the seller has not added one.

7. External Interface Requirement

7.1 Data Interfaces

The application stores users, listings, favorites and status information in a MySQL 8 relational database on the same host. No external data feed or third-party API is required for version 1. Test seed data is loaded from SQL scripts kept in the project repository.

7.2 User Interfaces

A simple responsive web interface usable from current Chrome and Edge browsers on laptops and mobile phones. Main pages: Home/Browse, Listing Detail, Publish/Edit Listing, Favorites, My Listings, Login/Registration and Profile. All forms use inline validation and English labels.

7.3 Other Interfaces

No payment gateway, e-mail/SMS service, logistics API or university identity API is required in version 1. The only communication interface is HTTPS between the browser and the server.

8. Non-Functional Requirements

8.1 System Performance

Req. Id	Description	Response Time
PERF-1.1	Time experienced by the user for regular operations such as moving between fields or pages of the same screen.	1 second
PERF-1.2	Search, filter and sort on a seeded database of 5,000 listings (P95).	2 seconds
PERF-1.3	Publishing a listing, including validation and persistence.	3 seconds

8.2 Information Security

Req. Id	Description
SEC-1.1	Passwords must never be stored or transmitted in plain text; only salted hashes are stored, and all traffic uses HTTPS.
SEC-1.2	Personal data (name, student number, e-mail) is visible only to the owner; other users see only the display name on a listing.
SEC-1.3	A user can modify or delete only his or her own profile, listings and favorites; every modifying request is authorised server-side.
SEC-1.4	After 5 consecutive failed logins the account is locked for 10 minutes.
SEC-1.5	All user input is validated server-side; database access uses safe parameterised queries or framework-provided ORM/query methods to prevent injection.

8.3 Availability

Req. Id	Description
AVA-1.1	The system is available during normal campus hours (08:00-24:00) on all teaching days.
AVA-1.2	Planned maintenance is announced at least 24 hours in advance and does not exceed 2 hours per week.
AVA-1.3	Unplanned downtime should not exceed 4 hours per month during the demonstration period.
AVA-1.4	After a crash, the system restarts with no loss of committed data (database transactions).

8.4 Capacity

Req. Id	Description
CAP-1.1	Year ONE data volume: up to 5,000 listings and 500 registered users.
CAP-2.1	Year ONE concurrency: at least 50 concurrent users searching without violating PERF-1.2.
CAP-3.1	Per-user limits: up to 200 favorites and 50 active listings per user.

Note on capacity planning: Year TWO and Year THREE volumes are out of scope for version 1. The platform is delivered as a course prototype sized for the Year ONE figures in CAP-1.1 and CAP-2.1; growth beyond these figures would be re-planned in a later version if the service is adopted by more students.

9. Domain Requirements

Req. Id	Description
DOM-1.1	All prices are expressed in MOP (Macau Pataca) with two decimal places; no currency conversion is provided.
DOM-1.2	The system serves only the MUST campus community; the item category list reflects typical campus goods (textbooks, electronics, household items, sports gear, others).
DOM-1.3	The transaction status vocabulary is fixed to Available / Reserved / Sold; payment, negotiation and physical handover are carried out offline and are out of scope.
DOM-1.4	Personal data is handled according to the Macao SAR Personal Data Protection Act (Law No. 8/2005): data is used only for the trading function and is deleted with the account.
DOM-1.5	Users should not publish illegal, dangerous or course-inappropriate items; the prototype states this rule even though automated moderation is not implemented.
DOM-1.6	Contact information is optional and is displayed only when a seller chooses to provide it.

10. Requirement Elicitation Process

This section records the actual situation at the time this revised document was prepared. No requirement is attributed to a person or group that has not provided feedback.

10.1 Method 1 - Assigned Customer Group (compulsory)

According to the course arrangement, Group 10 is the assigned customer group of Group 11; Group 11 in turn acts as the customer of Group 12. Group 11 contacted Group 10 and received feedback on 2026/10/20. Two issues were raised. First, the previous draft did not clearly define what should happen to related records when a listing or account is deleted. In response, REQ-1.3 and REQ-2.2 now state that deleting a listing removes the favorites that reference it, and deleting an account removes the user profile, the user's own listings and favorites, together with favorites that point to those deleted listings. Second, Group 10 pointed out that the previous automatic-matching design did not define how one listing should be handled when it matched several want-to-buy requests. Since automatic buyer-seller matching is not necessary for the basic course prototype and would add extra rules and testing work, version 1 keeps automatic matching outside the project scope.

10.2 Method 2 - Real Potential Users (optional)

This optional method has not been carried out yet, so this version does not claim that external students were interviewed. The team may conduct short interviews later if time permits and will only include findings that were actually collected.

10.3 Method 3 - Team Members as Surrogate Users (optional)

Because the team members are also university students and potential users of a campus second-hand platform, the group discussed what functions would be necessary for a simple first version. The discussion focused on functions that are useful and feasible within a third-year course project:

- A seller should be able to publish and manage a product listing easily.

- A buyer should be able to browse, search and filter products.
- Users should be able to save interesting products as favorites.
- Sellers should be able to mark items as Reserved or Sold to avoid repeated enquiries.
- Payment and handover should remain outside the system to keep the project manageable.
- Advanced AI matching, online payment, built-in chat and external verification should not be included in version 1 because they would significantly increase implementation and testing complexity.

10.4 Requirement Source Summary

Source	Current Status	Effect on This Version
Assigned customer group (Group 10)	Contacted; feedback received on 2026/10/20.	Used to clarify deletion-related data handling (REQ-1.3, REQ-2.2). Automatic matching kept outside version 1.
Real potential users	Not conducted yet; optional.	No interview findings are claimed.
Group 11 team discussion	Completed.	Used to define the realistic baseline requirements in Sections 5-9.

11. Contributions to This Report

This report was produced jointly by the four members of Group 11. Each member contributed approximately 25% of the total work.

Member	Student No.	Contribution to This Report	Share
Wu Yutong	1240016186	Led the team; organised the elicitation contact with Group 10; wrote Sections 1, 3 and 4 (Introduction, Constraints, Assumptions); compiled and finalised the complete report.	25%
Wang Yike	1240012305	Wrote Section 2 (General Description) and the detailed requirements REQ-1.1 to REQ-3.1; maintained the master list in Section 5.	25%
Chen Moxiong	1240008865	Wrote the detailed requirements REQ-4.1 and REQ-5.1 and Section 8 (Non-Functional Requirements).	25%
Gao Shengzhe	1240008949	Wrote REQ-6.1 and Sections 7, 9 and 10 (Interfaces, Domain Requirements, Elicitation Process); applied the template formatting and proofread the final document.	25%